

Anurag Shukla

Software Engineer | Full-Stack & Applied AI

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📍 Prayagraj, India

SKILLS

Languages: Python, JavaScript/TypeScript, Java, Typescript

AI/ML: Scikit-learn, Large Language Models (LLMs), XGBoost, OpenAI API, LangChain, Hugging Face

Frameworks: React, Next.js, TailwindCSS, Node.js, FastAPI, Convex, PostgreSQL, MongoDB

DevOps & Tools: Docker, AWS (EC2/S3), Git, Vercel, Postman

PROFESSIONAL EXPERIENCE

ECAVO AGRO DAILY PVT. LTD. [🔗](#), AI Agent Development Intern

06/2026 – Present

Technologies: Python, Raspberry Pi, Arduino, OpenCV

Prayagraj, India

- Built industrial automation solutions using Python, Raspberry Pi, Arduino, and camera systems to reduce manual work and improve production processes.
- Developed computer vision applications with OpenCV for real-time product monitoring and basic quality inspection on the production line.
- Integrated cameras, sensors, and hardware with Python scripts to automate machine operations and support smart factory workflows.

IBM [🔗](#), Software Engineering Intern (Full-Stack / AI)

06/2025 – 08/2025

Technologies: React 19, TypeScript, Convex, Gemini API, Clerk Auth, React Router v7

Prayagraj, India

- Architected a real-time dual-pane resume editor using React 19 + Convex reactive database — syncing multi-section form state (personal, experience, education, skills) to a live preview component with zero-latency, no polling or debounce.
- Integrated Gemini API to generate resume bullet points from job titles using structured prompting, reducing manual editing time for users.
- Engineered PDF export and public resume sharing pipeline — generating unique shareable URLs per resume and rendering print-formatted layouts via browser PDF API, enabling recruiter access without authentication.

UNITED COLLEGE OF ENGINEERING AND RESEARCH [🔗](#), Machine Learning Intern (Applied NLP)

06/2024 – 09/2024

Technologies: XGBoost, Scikit-learn, Streamlit, Pandas / NumPy, NLTK Vercel (Python)

Prayagraj, India

- Built an end-to-end NLP pipeline (text preprocessing → Count Vectorizer → MinMax scaling → XGBoost) achieving production-ready inference at 100–200ms per prediction on Amazon Echo review corpus.
- Designed a memory-efficient bulk processor using chunked Pandas reads (10k rows/chunk), enabling sentiment classification on 100k+ row datasets without memory overflow.
- Deployed serialized ML artifacts (XGBoost + CountVectorizer + MinMax scaler) on Vercel Python runtime — enabling zero-infrastructure live inference with a Streamlit frontend and public URL.

PROJECTS

AI Smart Traffic Navigator [🔗](#), YOLOv8, OpenCV, Python, Arduino, Streamlit

02/2026 – 04/2026

- Built real-time multi-class vehicle detection using YOLOv8 + OpenCV with PCU-based density scoring, feeding a dynamic signal algorithm computing adaptive green timings from density, wait time, and optical flow.
- Implemented emergency vehicle preemption and pedestrian-aware signaling with real-time override of standard traffic cycles.
- Engineered a multi-threaded pipeline integrating CV inference, decision logic, and Arduino hardware control via PySerial — with Pygame simulation fallback and live Streamlit monitoring dashboard.

AI AGENT BUILDER [🔗](#), Next.js 15, TypeScript, OpenAI SDK, React Flow, Convex, Clerk, Arcjet

08/2025 – 02/2026

- Engineered a visual drag-and-drop AI agent builder with React Flow + Next.js 15 App Router — enabling multi-step OpenAI agent workflow design (tool calls, prompt chains, decision branches) without writing orchestration code.
- Built a serialization layer converting React Flow node graphs into executable OpenAI agent workflows.
- Hardened with Arcjet bot protection + Clerk auth — rate limiting, bot fingerprinting, and per-user route isolation across Convex serverless backend.

EDUCATION

B.Tech, Computer Science & Engineering (AI/ML Specialization),

UNITED COLLEGE OF ENGINEERING AND RESEARCH

10/2023 – Present

Prayagraj, India

CERTIFICATES

Machine Learning Specialization — DeepLearning.AI / Stanford (Coursera)

Oracle Cloud Infrastructure 2025 — Certified AI Foundations Associate

Full-Stack Web Development — freeCodeCamp